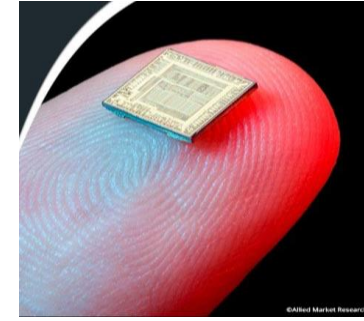
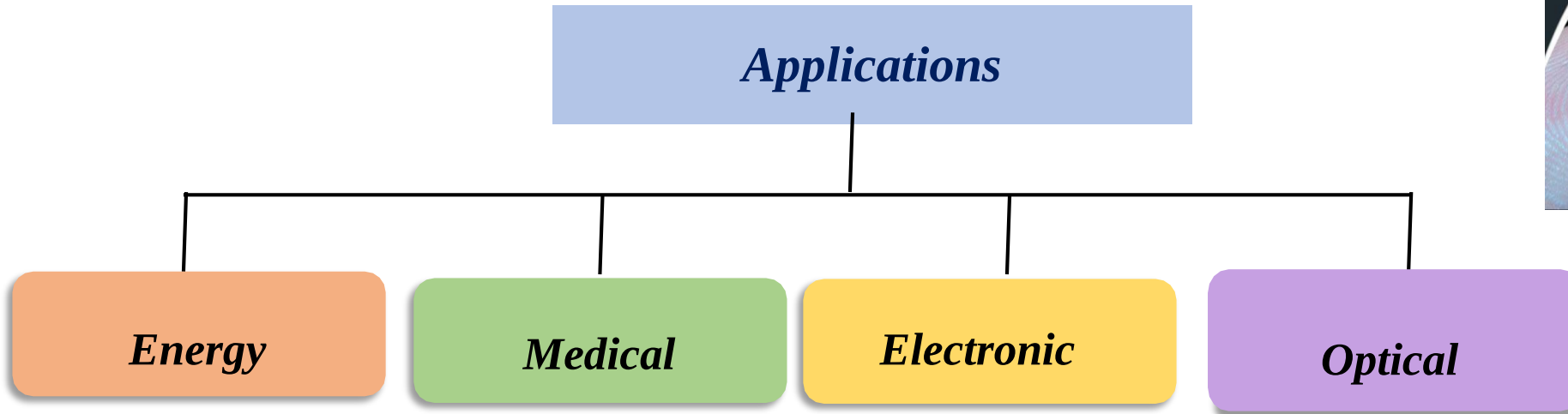


Applications of nanoscience and nanotechnology

Dr Nishad G Deshpande

IIIT Surat

Application of nanomaterials -

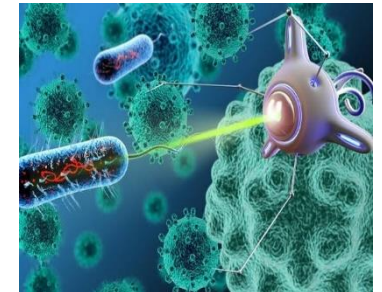


- ➡ **Supercapacitors**
- ➡ **Batteries**
- ➡ **Soler cell**
- ➡ **Fuel cell**
- ➡ **Wind turbines**

- ➡ **Medicines**
- ➡ **Drug delivery**

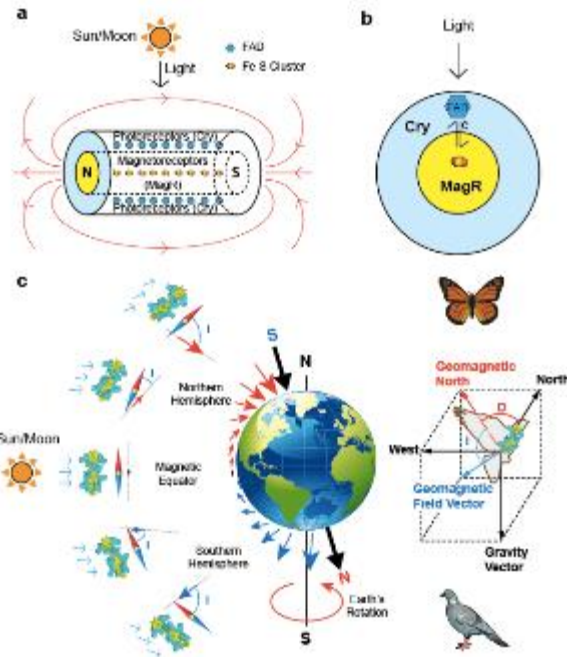
- ➡ **Memory chips**
- ➡ **Touch screen layers**
- ➡ **Electronic gadgets**

- ➡ **Optical sensors**
- ➡ **Imaging**
- ➡ **Optoelectronics**
- ➡ **Non-linearity**
- ➡ **Photonic**

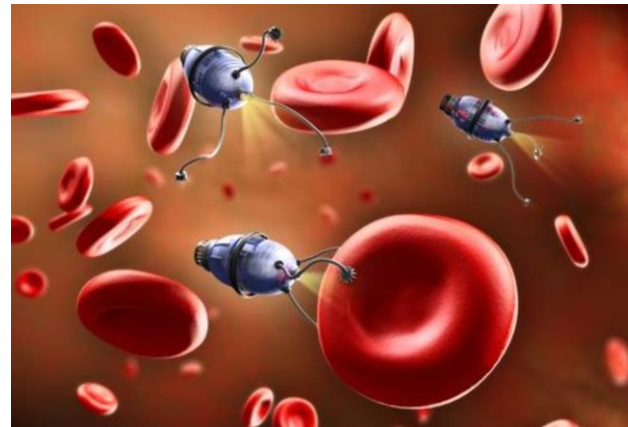


Magnetism in Medicine

- Nanomedicine uses “nano-tools” (e.g. **smart nanoparticles**) that are roughly **1000 times smaller than a cell** (knives to micro- to nano-surgery ...)



Synthesis of hemoglobin



Removal of metal (Iron) splinters, shrapnel, etc.



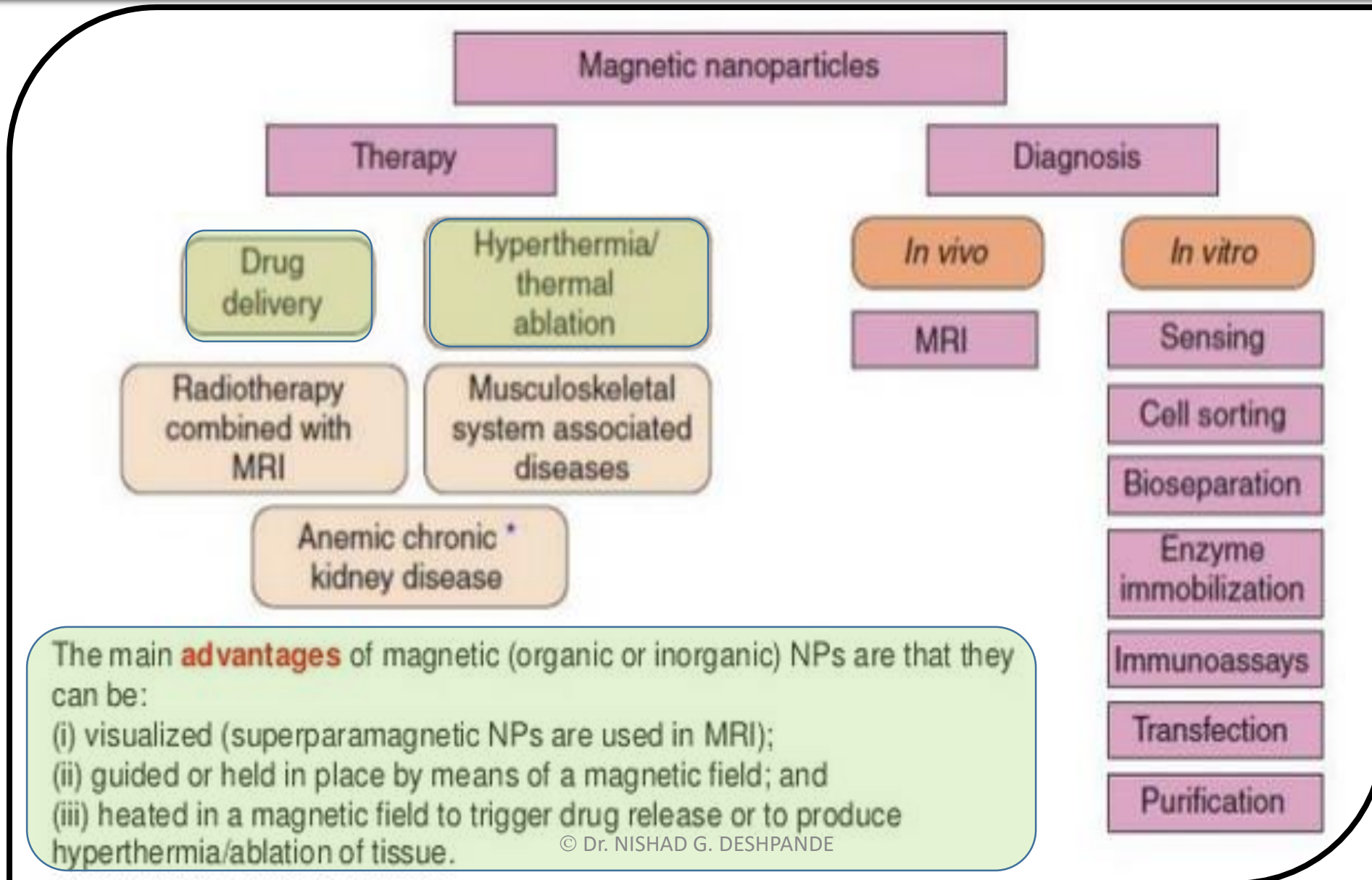
“This certainly is a quick way of removing steel splinters, doctor!”

Animals use magnetic fields to navigate

• Many other applications

- Holding prosthetics
- Guiding instruments through the body
- MRI

Magnetic Nanoparticles for various Bio-Medical Applications



Environmental Applications

1. Sensors for monitoring environmental pollutions-Gas sensors
2. Nanomaterials for wastewater/contaminated water treatments

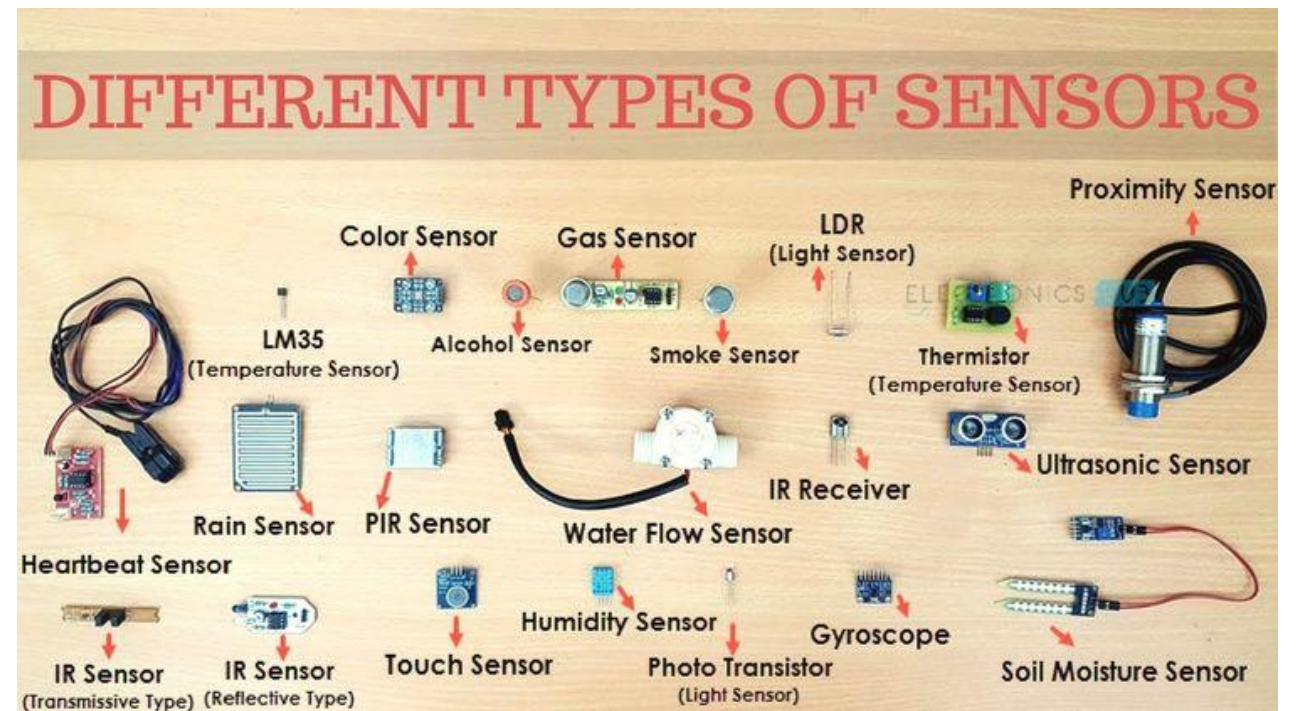
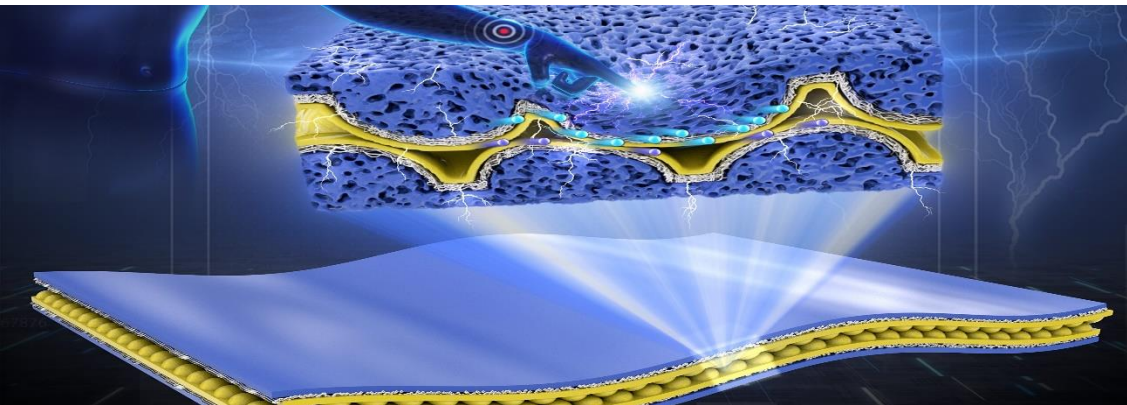
Environmental Pollution Remedies-

Air pollution

DEFINITION :-

Nanosensors are nanoscale devices that measure physical quantities and convert these to signals that can be detected and analyzed

- There are different types of **nanosensors** in the market and in development for various applications, most notably in defense, environmental, and healthcare industries.
- These sensors share the same basic workflow: a selective binding of an analyte, signal generation from the interaction of the nanosensor with the bio-element, and processing of the signal into useful metrics.

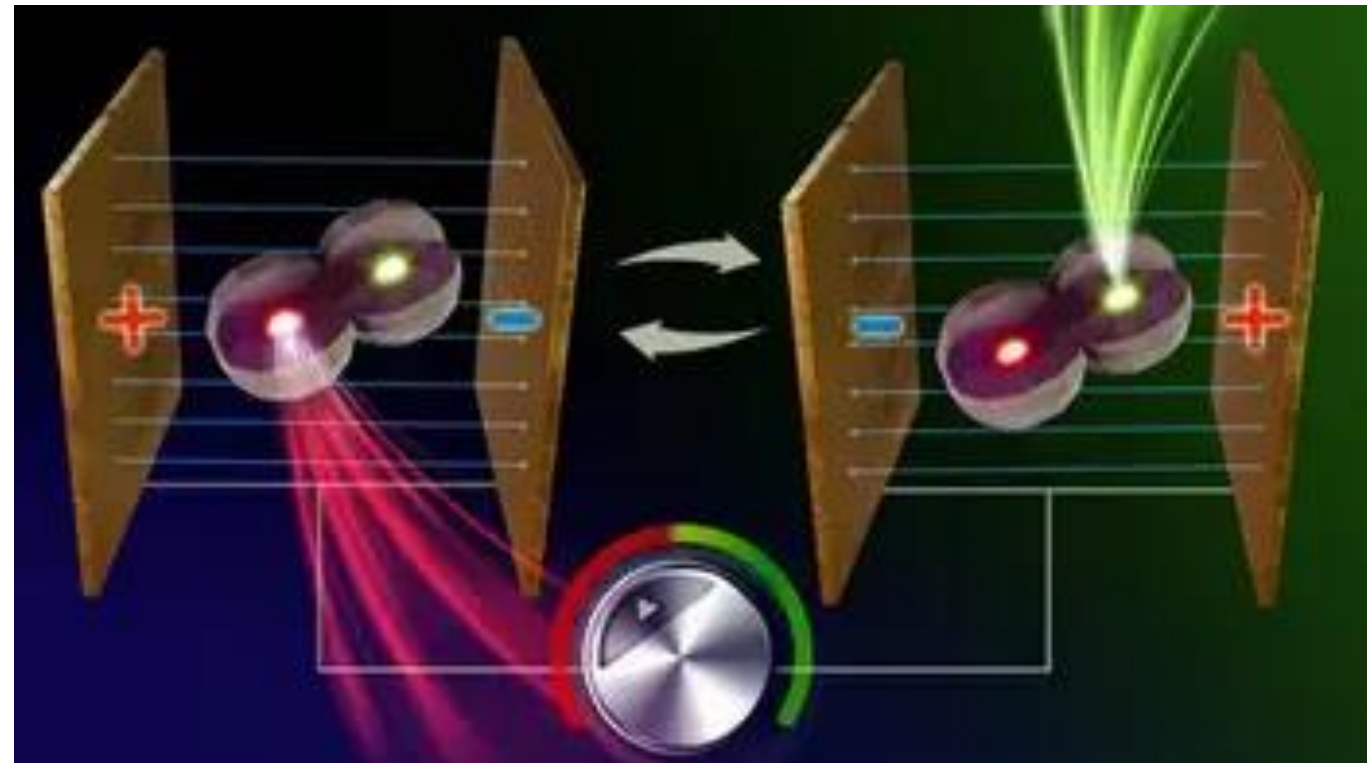
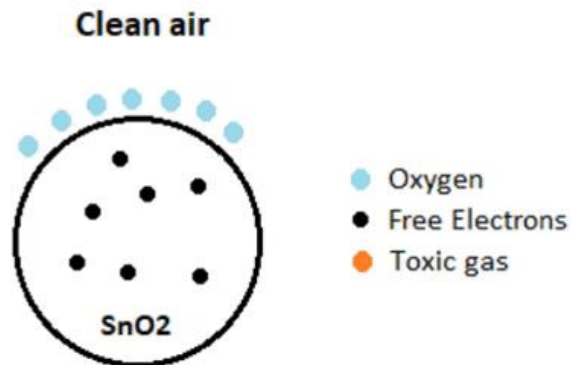


CHARACTERISTICS:-

- ✓ Nanosensors can have increased **specificity** because they operate at a similar scale as natural biological processes, allowing functionalization with chemical and biological molecules, with recognition events that cause detectable physical changes.
- ✓ Enhancements in **sensitivity** stem from the high surface-to-volume ratio of nanomaterials, as well as novel physical properties of nanomaterials that can be used as the basis for detection, including nanophotonics.
- ✓ Nanosensors provide real-time monitoring compared to traditional detection methods such as chromatography and spectroscopy

Applications:-

- ❖ Defence and military
- ❖ Food and the environment
- ❖ Healthcare



Different Types of Gas sensors

Gas sensors are typically classified into various types based on the type of the sensing element it is built with.

Below is the classification of the various types of gas sensors based on the sensing element that are generally used in various applications:

Metal Oxide based gas Sensor.

Optical gas Sensor.

Electrochemical gas Sensor.

Capacitance-based gas Sensor.

Calorimetric gas Sensor.

Acoustic based gas Sensor.

Gas Sensor Construction

Of all the above-listed types, the most commonly used gas sensor is the Metal oxide semiconductor based gas sensor. All Gas sensors will consist of a sensing element which comprises of the following parts.

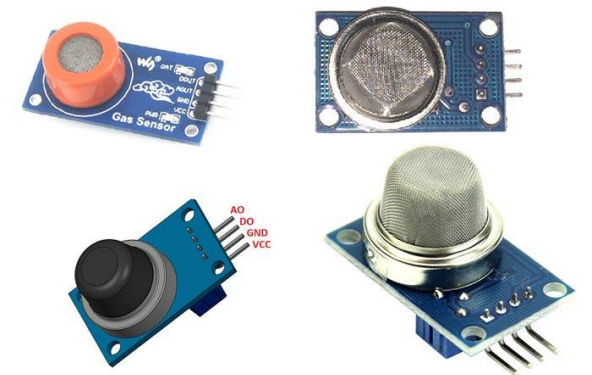
Gas sensing layer

Heater Coil

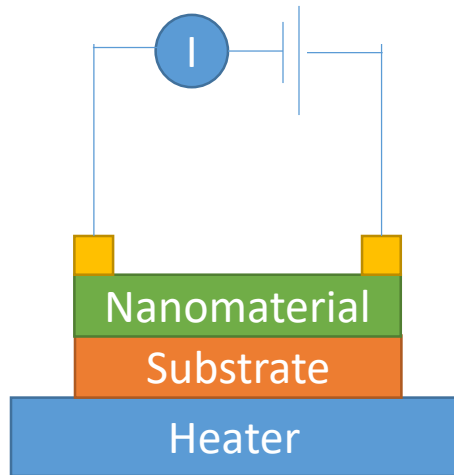
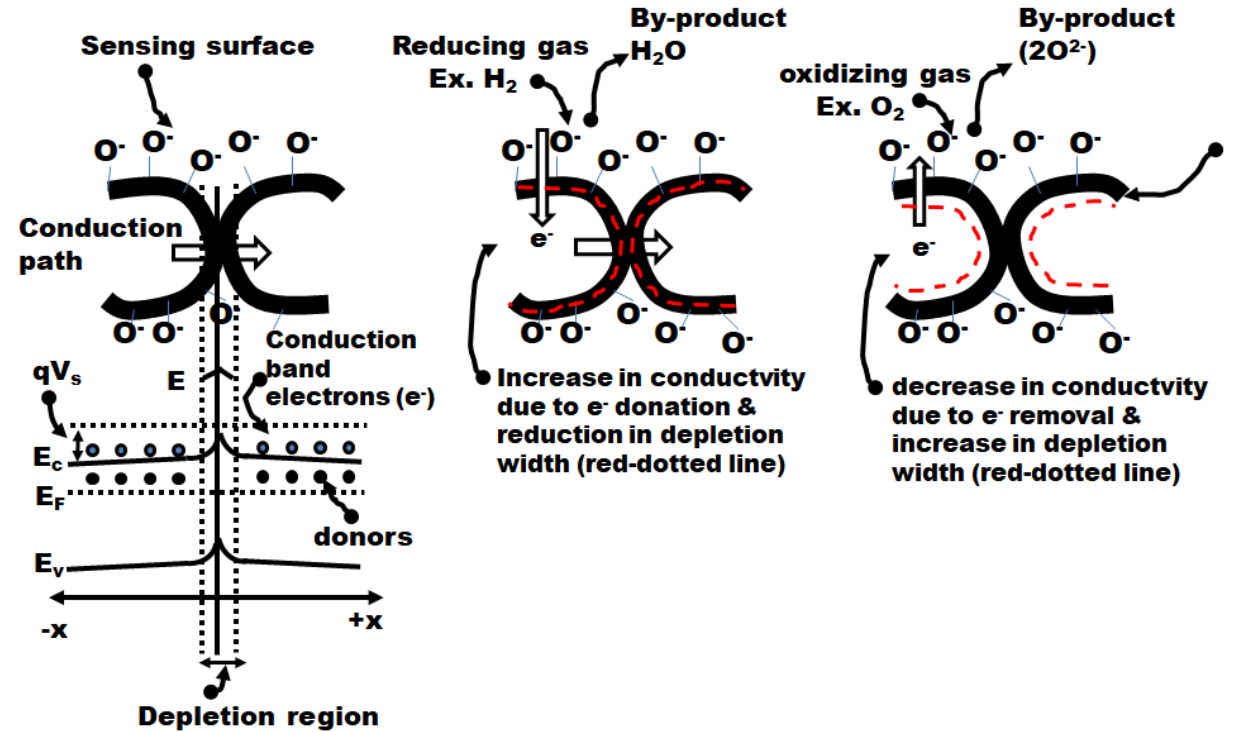
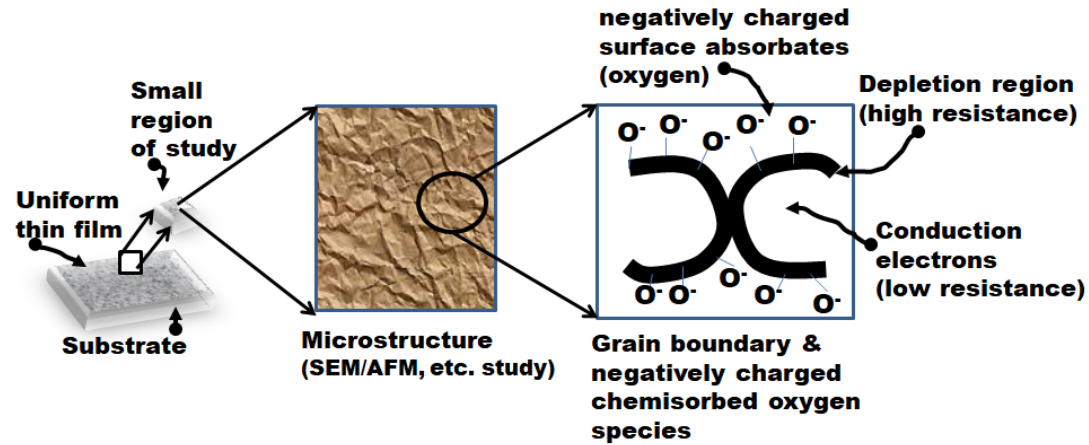
Electrode line

Tubular ceramic

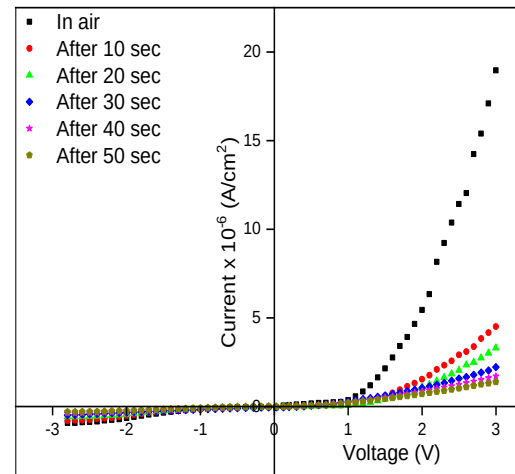
Electrode



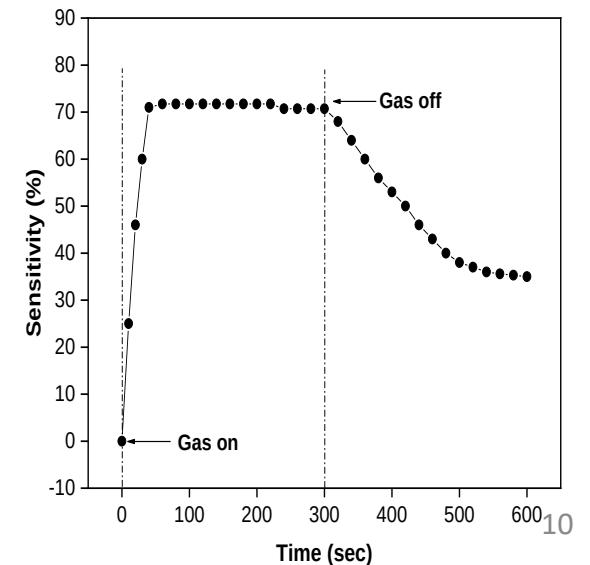
Gas Sensor

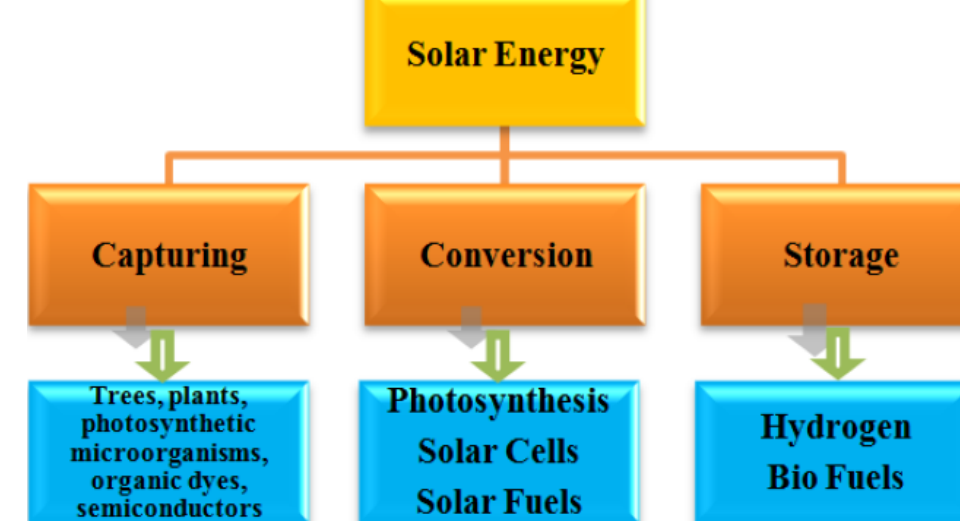
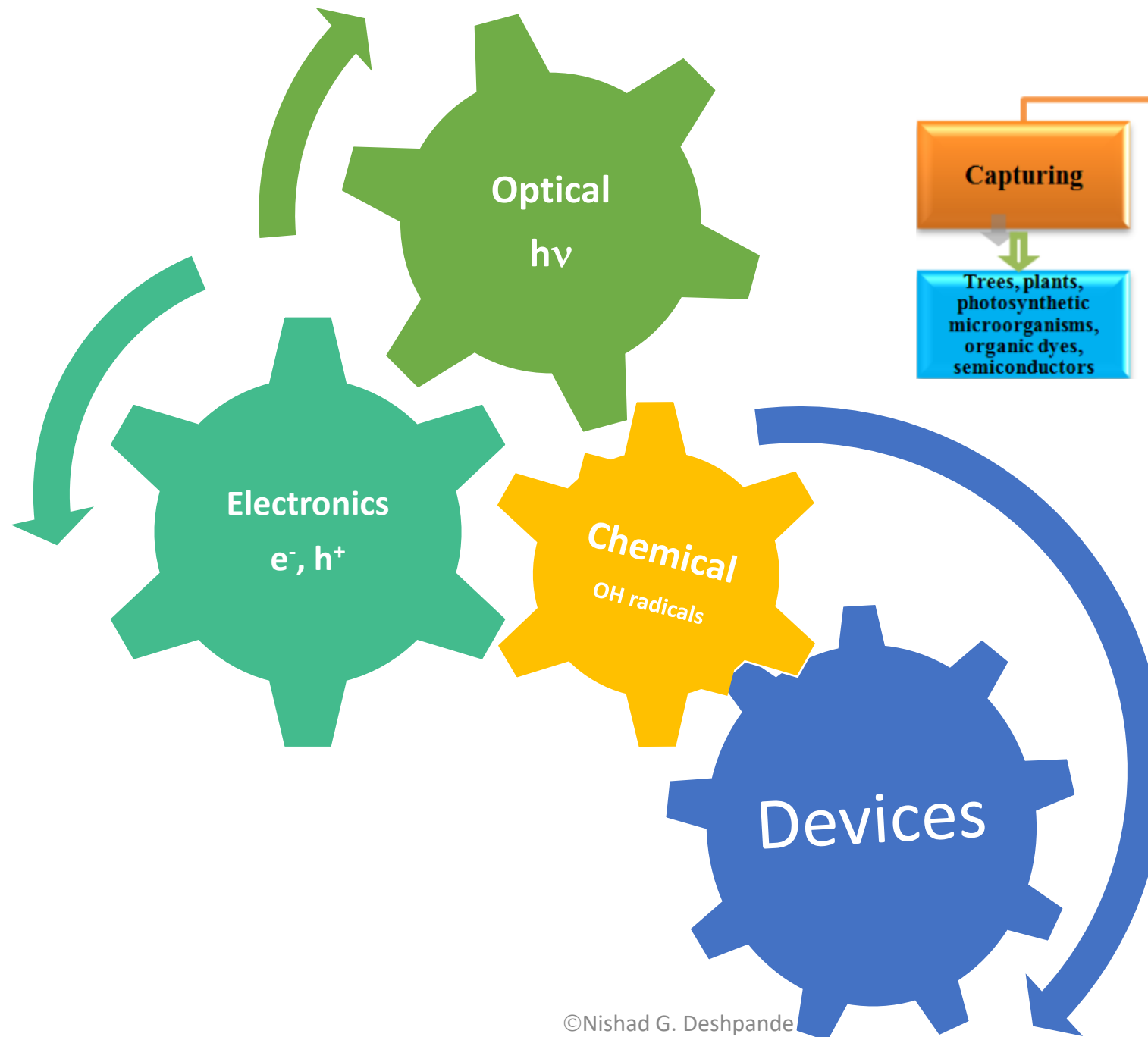


Metal Contacts



Nishad G. Deshpande

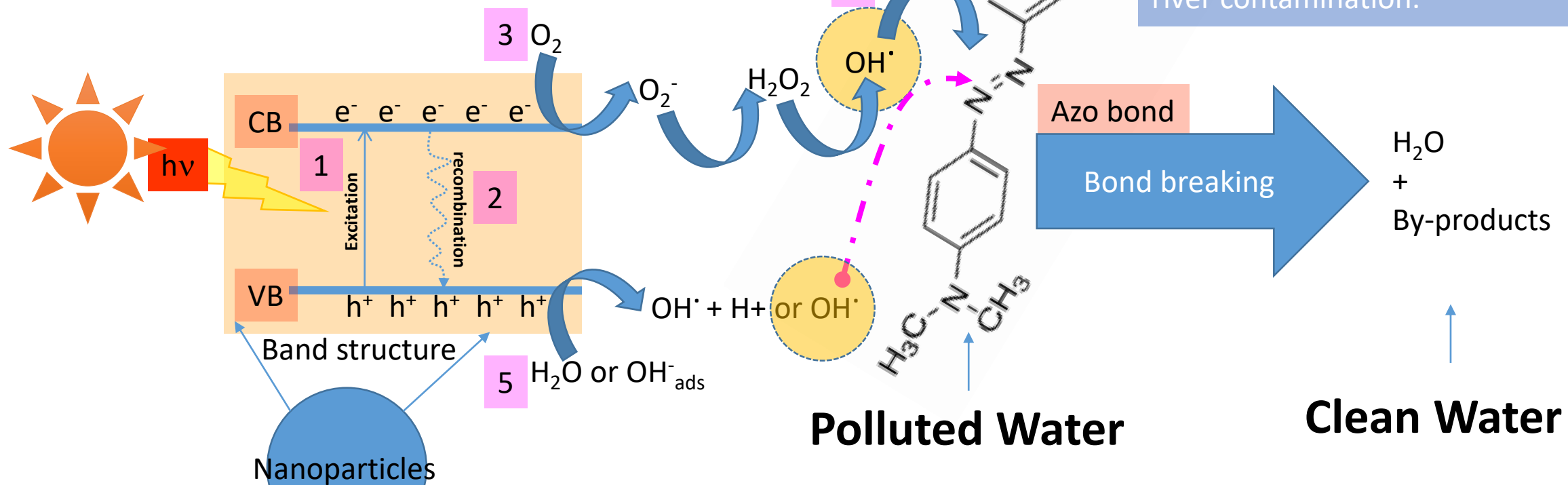




Environmental Pollution Remedies-

Water pollution

1. Excitation of semiconductor (nanoparticle) by Light
2. Recombination of Charge carriers
3. Transfer of excited e^- to dissolved O_2
4. Reduction of dye by hydroxyl radicals
5. Oxidation of H_2O and OH^- by holes



Most toxic dyes from textile industries are discharged into rivers, leading to significant water pollution. Pharmaceutical and chemical industries also contribute to river contamination.

Photocatalytic dye degradation for wastewater treatment using nanoparticles



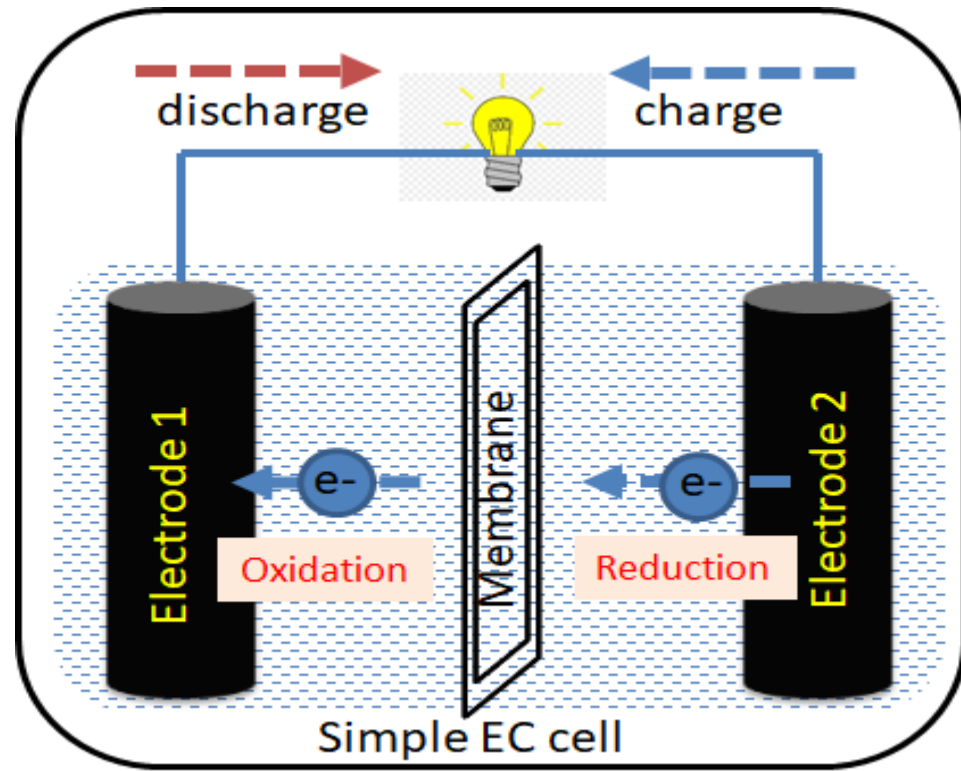
Energy devices

1. Electrochemical devices-Hydrogen Fuels
 - Supercapacitors
 - Battery

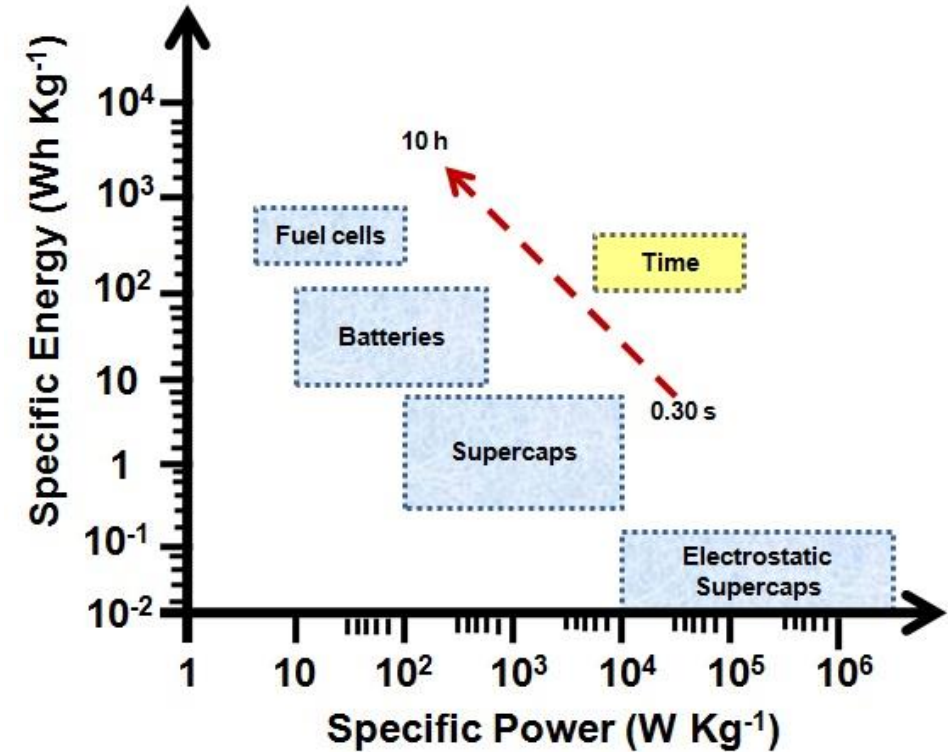
Different electrochemical energy storage and conversion devices

EC ENERGY STORAGE & CONVERSION DEVICES				
CAPACITORS	BATTERIES		FUEL CELLS	
ELECTRICAL DOUBLE LAYERED CAPACITORS (EDLC)	RECHARGABLE BATTERIES	LEAD-ACID	METAL AIR	
		NICKEL BASE		
		ZINC BASE		
		SODIUM BASE		
		METAL/AIR		
PSEUDOCAPACITORS (PC)		HIGH TEMPERATURE OR AMBIENT TEMPERATURE LITHIUM	HYDROGEN	DIRECT
				INDIRECT
HYBRID CAPACITORS (HC)	FLOW BATTERIES	REDOX FLOW	MICROBIAL	MEDIATOR
		HYBRID FLOW		MEDIATOR FREE

Electrochemical cell and its characteristics



Schematics of simple electrochemical cell



Energy density, power density and delivery timescale for different energy storage and conversion devices

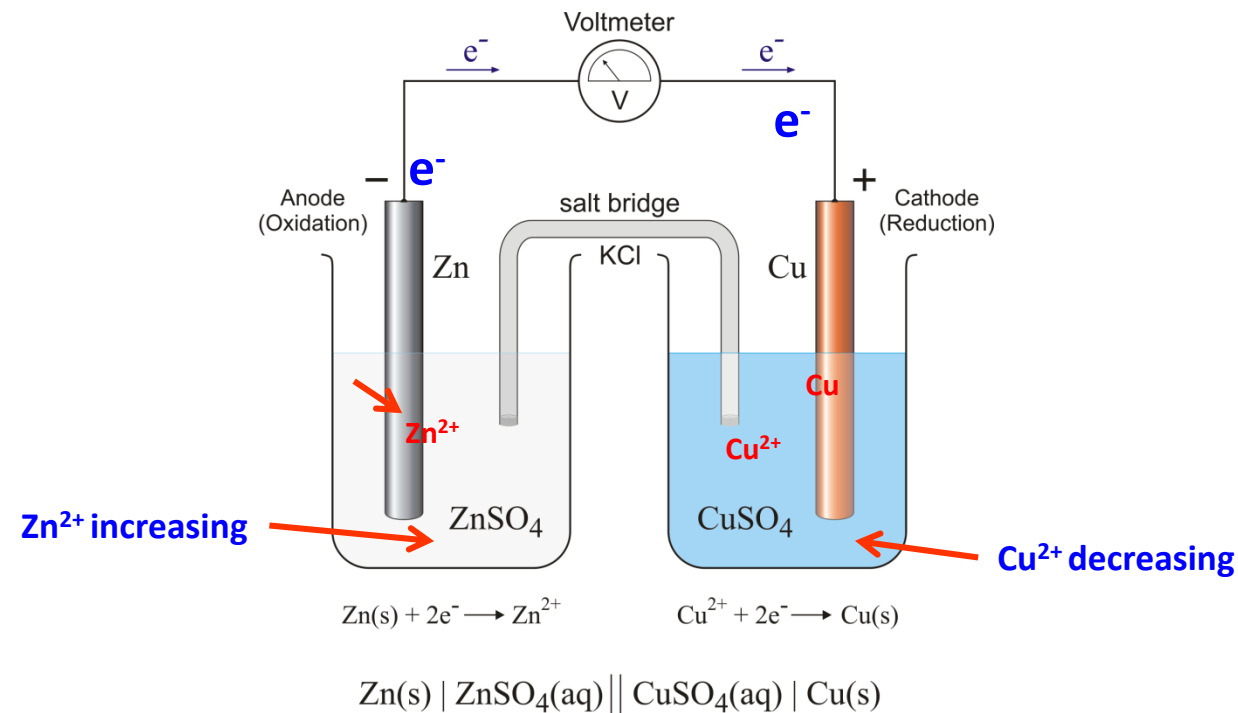
Introduction to battery and its Type

Batteries

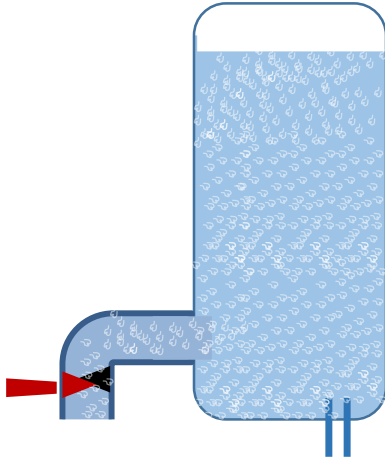
- ✓ The energy device which stores electrical energy in the form of chemical energy.
- ✓ Provides electrical energy as per requirement.
- ✓ Hold limited capacity and has limited life span.

Basic Design

- ✓ Simple cell

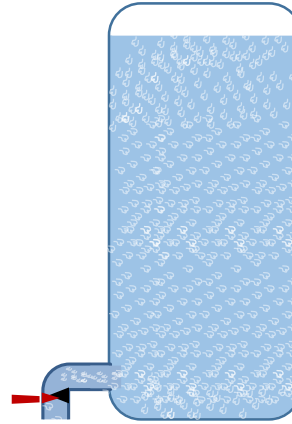


Battery Analogy



High Power consumption

- fast discharge
- used for Recording camera, drilling machine etc.
- Need rechargeable battery



Low Power consumption

- slow discharge
- used for Kitchen clock, Calculators etc.
- Need Long Lasting Single use battery

Primary Batteries
(Non-rechargeable)

Secondary Batteries
(Rechargeable)

Battery Types

Primary Batteries (Non-rechargeable)

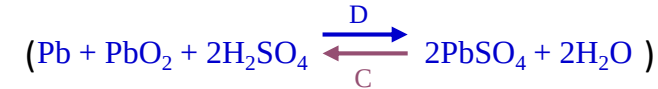
- ✓ Carbon-zinc dry cells (1.5 V)
- ✓ Alkaline cells (Zn-MnO₂)
- ✓ Zinc chloride cells
- ✓ Lithium (metallic) battery



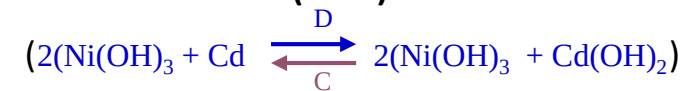
- useful in small electronic devices, toys, torches etc.

Secondary Batteries (Rechargeable)

- ✓ Lead-Acid Batteries



- ✓ Nickel-Cadmium (NiCd) Batteries

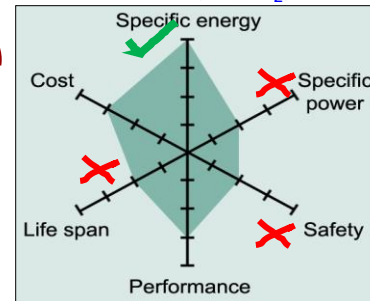


- ✓ Lithium ion battery (intercalated Lithium compound)

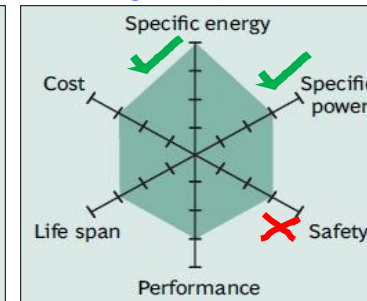
- useful in mobile phones, Laptops, Electric cars etc.

Ion batteries-

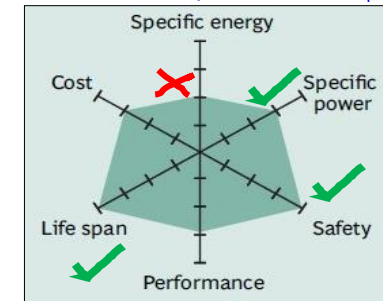
Li-Cobalt (LiCoO₂)



Nickel-Manganese-Cobalt (NMC)



Lithium Phosphate (LiFePO₄)



Lithium Polymer Batteries

- uses solid electrolyte
- 50° – 60° to attain conductivity

	Nominal Scale	Actual scale OCV
Lead Acid	2V/ Cell	2.10V/Cell
NiCd, NiMH	1.20V/Cell	1.25V/Cell
Li-ion	3.60V/Cell	3.70V, 3.80V/Cell *

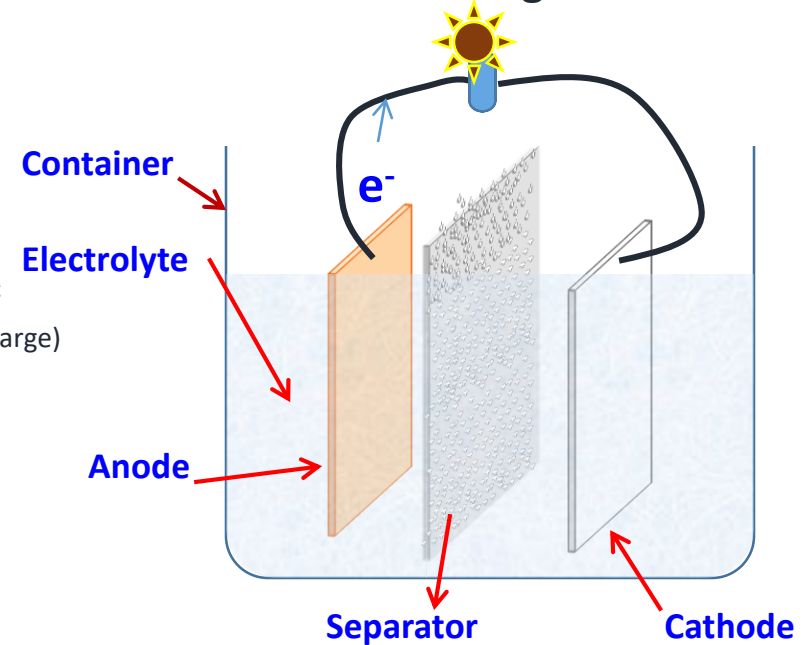
*Cathode material affect OCV. Manganese raises voltage.

Rechargeable batteries and their environmental issues

Components in a Battery

- ✓ **Cathode** (release electrons to circuit)
- ✓ **Anode** (collect electrons from external circuit)
- ✓ **Electrolyte** (medium for ionic transfer and chemical reaction)
- ✓ **Separator** (a permeable membrane; usually polymeric allowing transport of ionic charge carries but avoid electrode contacts due to dendrite growth in charge –discharge)
- ✓ **Container** (rigid, non-conducting battery case)

Basic Design



Choice of Anode and Cathode materials

Li → 3.04 V ← High Electrochemical Potential

Mg → 2.37 V

Al → 1.66 V

Zn → 0.76 V

Fe → 0.44 V

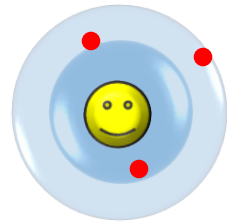
H → 0 V

Hg → -0.24 V

Cu → -0.34 V

Ag → -1.69 V

F → -2.8 V



↓
Li

Highly reactive and unstable

✓ LiCoO₂ (Lithium ion/polymer batteries)

✓ LiFePO₄,

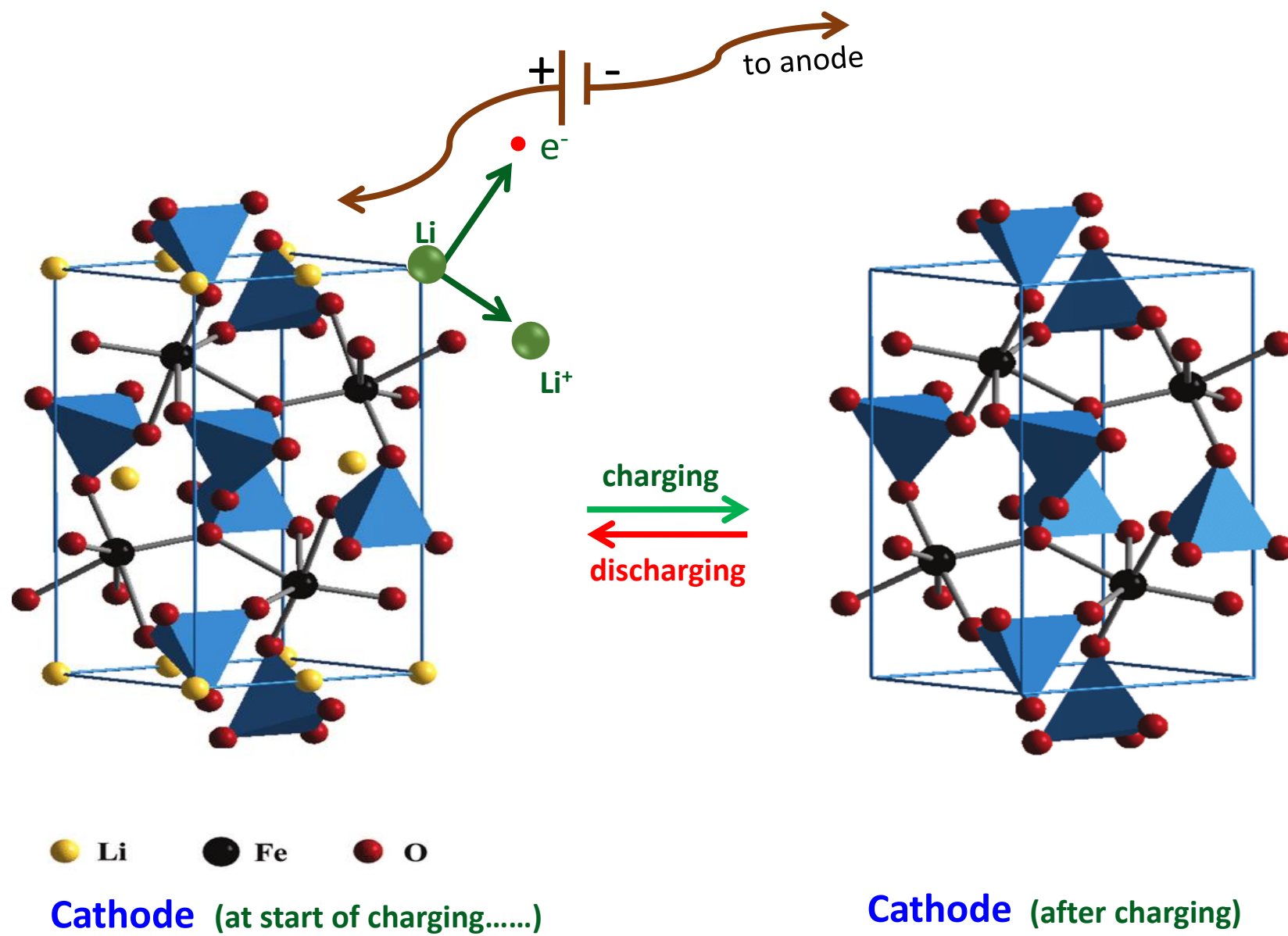
✓ LiMn₂O₄,

✓ Li₂MnO₃,

✓ LiNiMnCoO₂

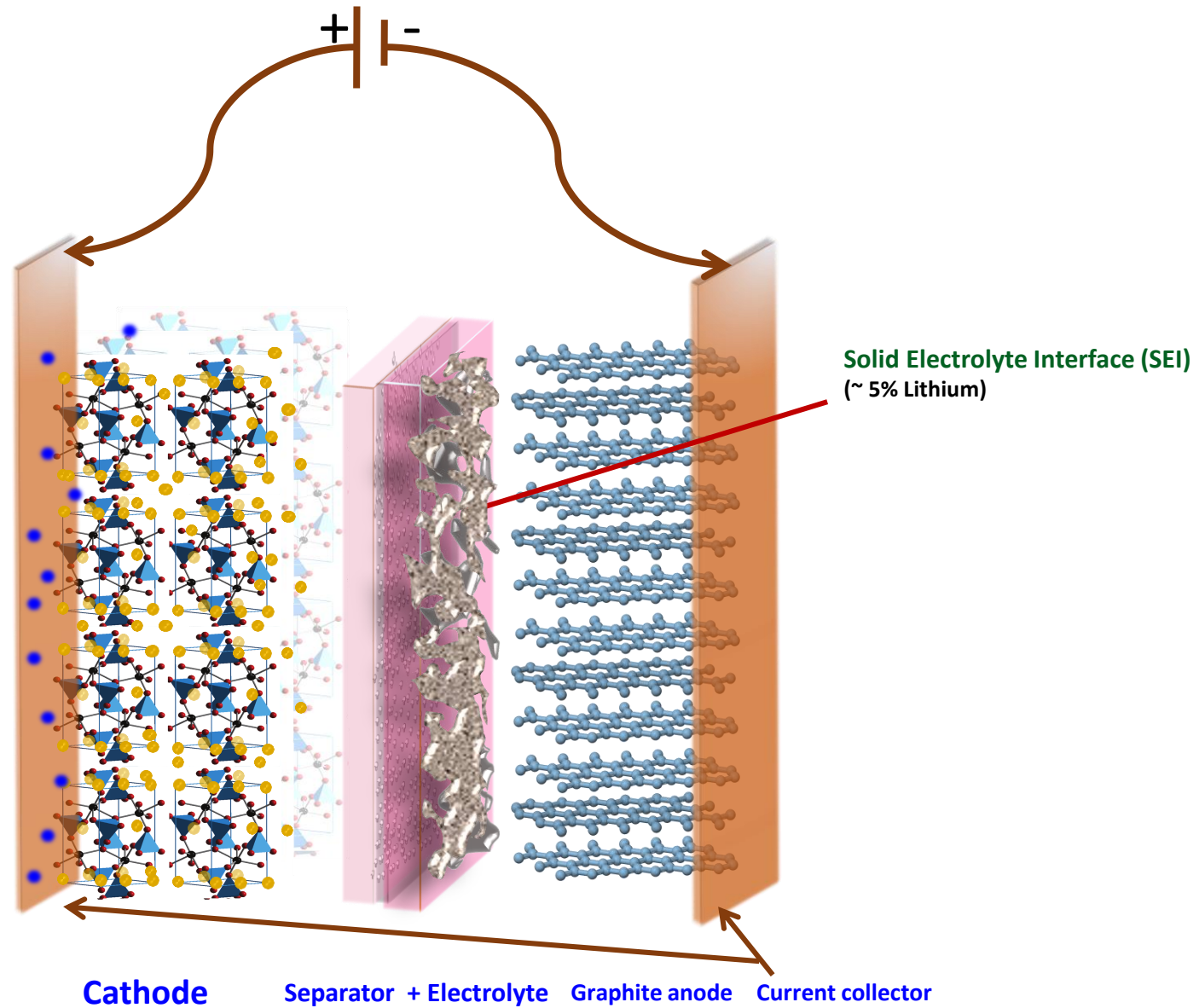


3.2 V Rechargeable Li-ion battery



Lithium ion battery: Working

Charging by External DC Load



Hazardous to human and environment

- Though Lead-Acid, NiCd and Lithium-ion based batteries are widely used in our daily electronics, these are hazardous to environment due to its high toxicity. So, it is proper time to replace these sources by non-toxic materials.
- Even small defect or overheating in Li-ion battery causes blast of battery and also Lithium is not abundantly available element.

Eco-friendly battery sources

✓ Sodium-ion Battery

✓ Zinc-air / Zinc-ion battery

Non-toxic material, low cost and relatively safe

(in primary research stage...)

The unintended presence of hydrogen, which leads to degradation of battery electrodes, is behind the short lifetimes of sodium-ion batteries, according to new research.

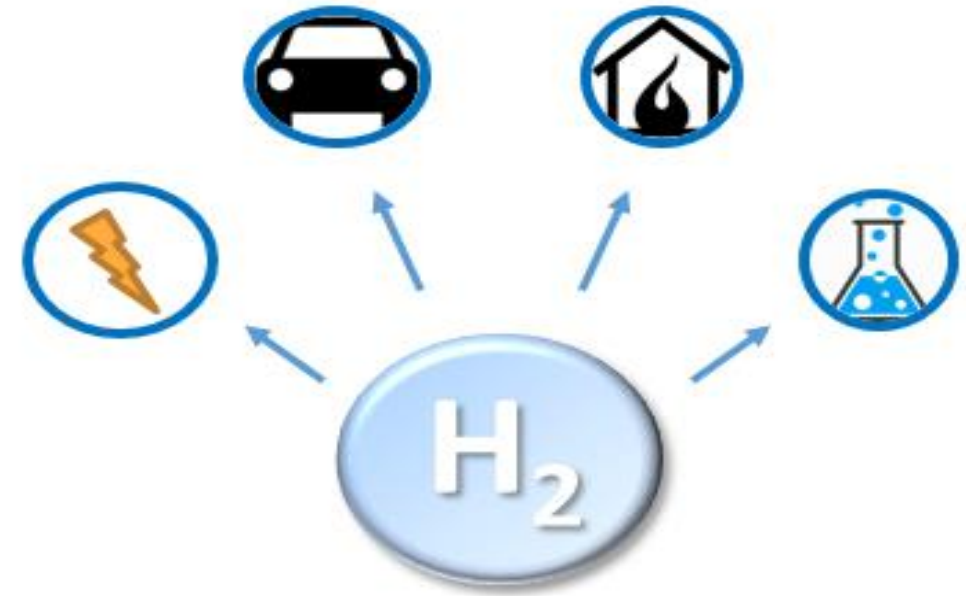
Zinc-air / Zinc-ion battery

Application of hydrogen:

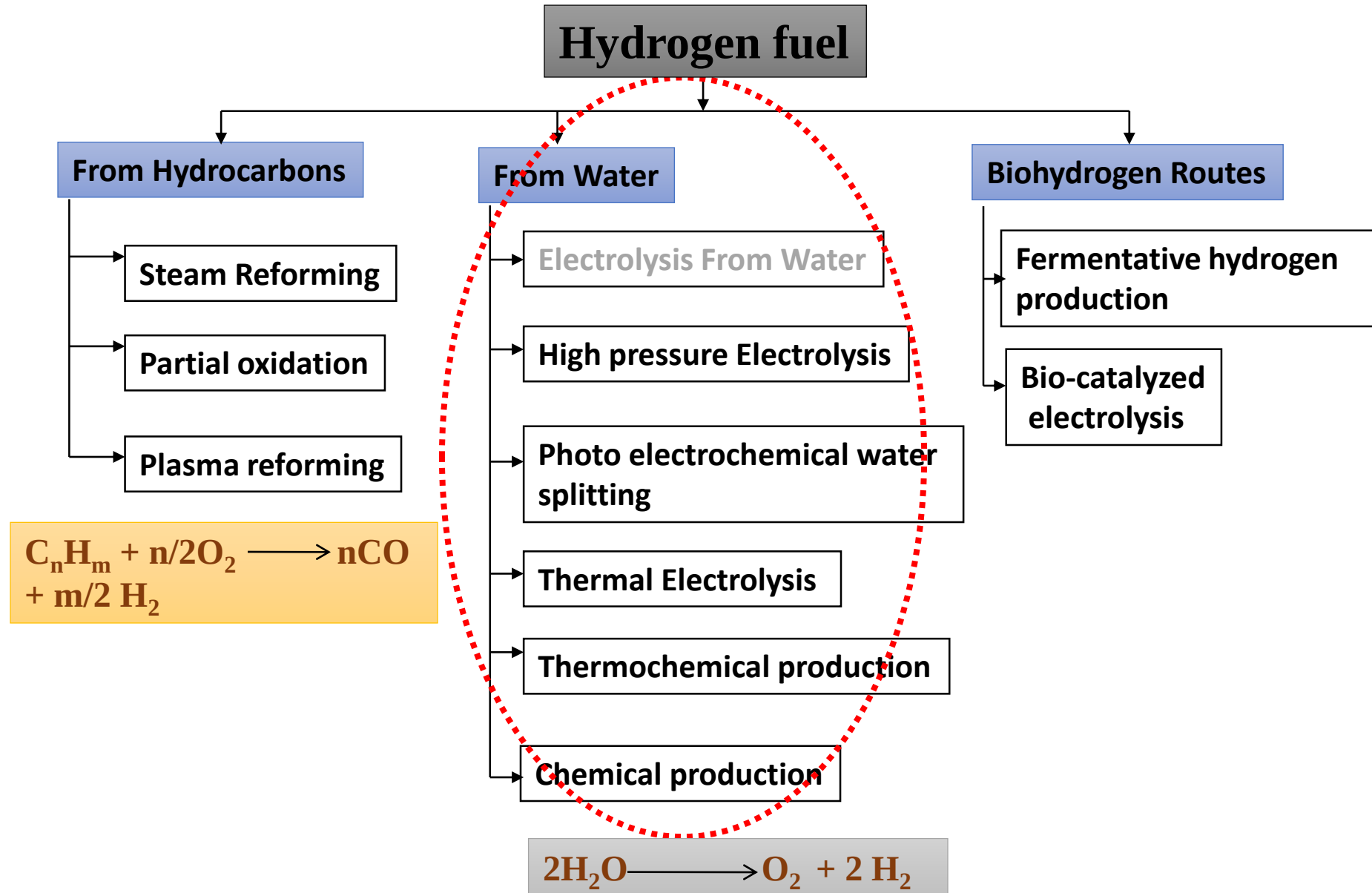
WHY HYDROGEN Fuels ?

- Hydrogen is the ultimate fuel for the future/next generation.
- It is clean - no greenhouse gases.
- Improved energy efficiency and abundant in nature.

- 📖 Produce electricity
- 📖 Cook Food
- 📖 Drive car
- 📖 Run Factories
- 📖 Personal Use
- 📖 Jet plane and Space craft



Preparation method of hydrogen



WATER-SPLITTING?

Water splitting is the general term for a chemical reaction in which water is separated into oxygen and hydrogen.

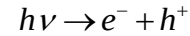
This can be achieved through following ways:

- **Electrolysis**
- **Thermal decomposition of water**
- **Photobiological water splitting**
- **Photo-electro-chemical water splitting**

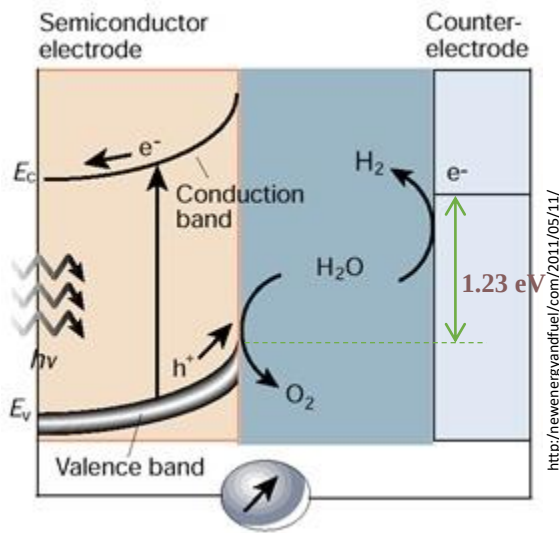
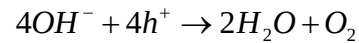
There are several points, which make PEC water splitting a promising technology for H₂ or O₂ generation:

1. Based on renewable sources-solar energy and water. Both are abundant.
2. Environmentally benign- no undesirable byproducts. No pollution issues.
3. Can be used on both large and small scale H₂ and O₂ production.
4. Integration of processes of solar light absorption to EHP generations and water splitting in a single photocell is possible.

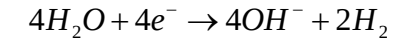
1) Absorption of light near the surface of the semiconductor creates electron-hole pairs.



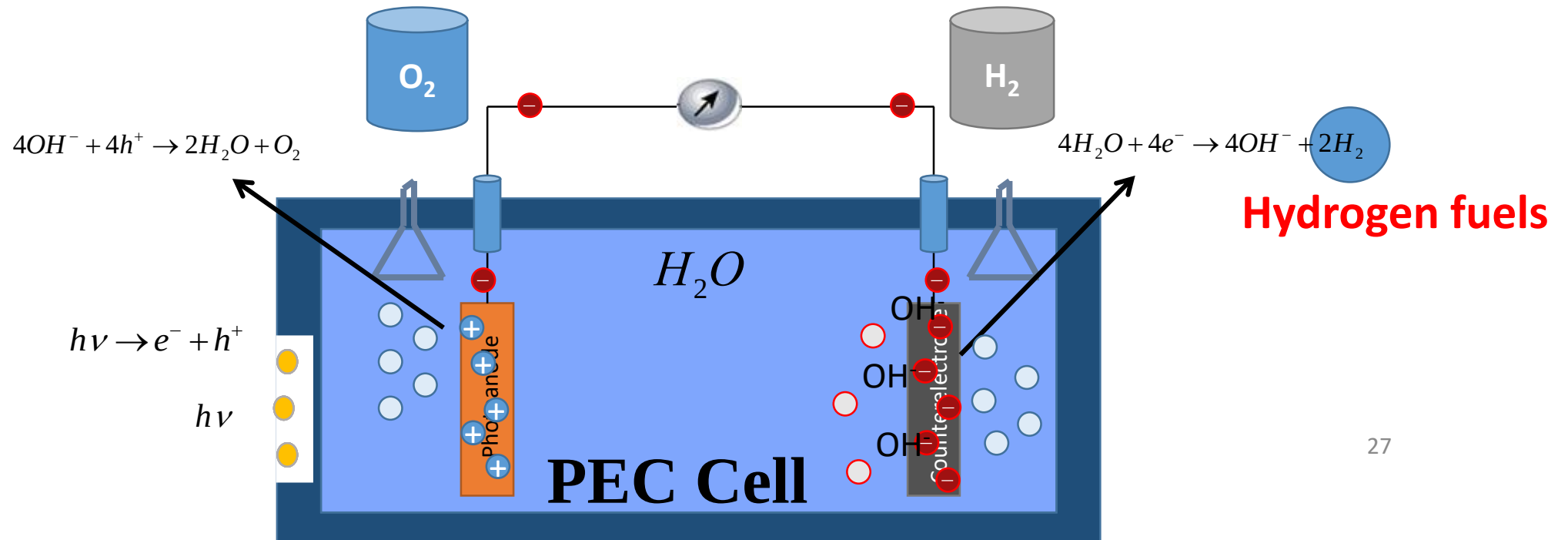
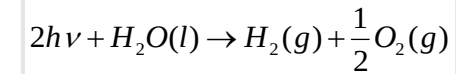
3) Holes (minority carriers) drift to the surface of the semiconductor (the photo anode) where they react with water to produce oxygen:



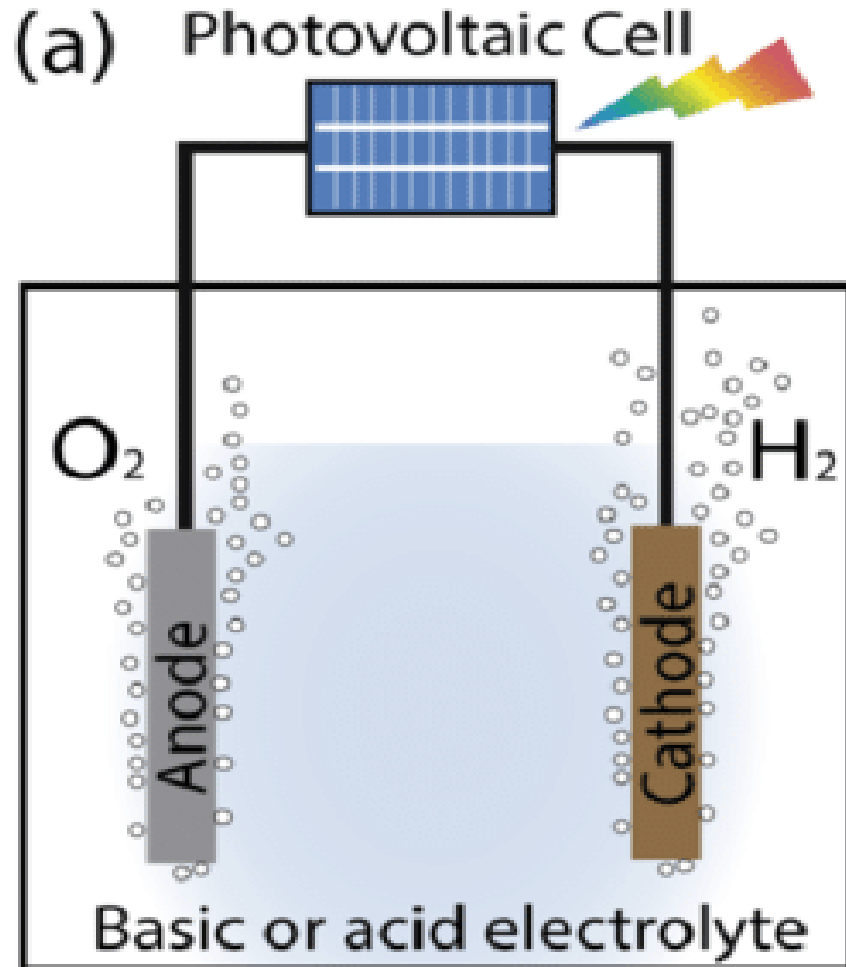
2) Electrons (majority carriers) are conducted to a metal electrode (typically Pt) where they combine with H^+ ions in the electrolyte solution to make H_2 :



4) Transport of H^+ from the anode to the cathode through the electrolyte completes the electrochemical circuit.



Photovoltaic electrochemical cell



Factor effecting on PV-assisted PEC cell

- @ nature of catalyst.
- @ nature of electrolyte.
- @ pH of electrolyte.
- @ large surface area.
- @ catalyst having long term stability.

Limitation for Hydrogen Fuel:

- ❑ Price of Hydrogen production.
- ❑ Storage problem.



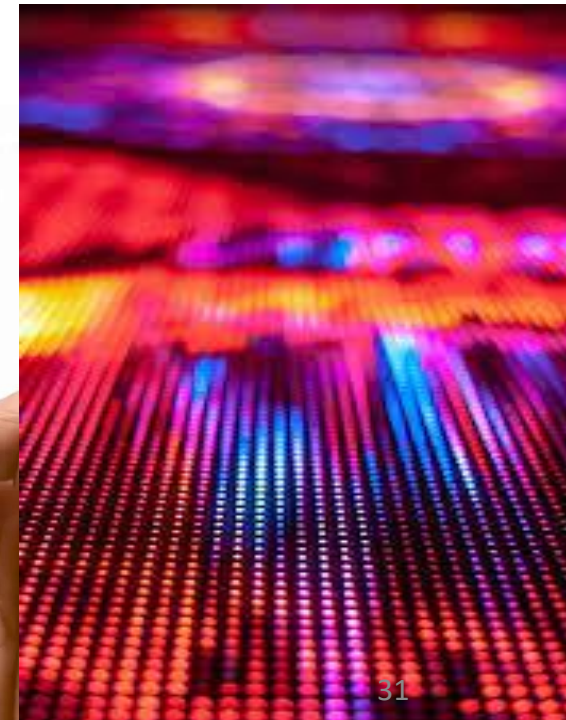
How to eliminate:

- ✓ Reducing the production cost.
- ✓ Improving efficiency.
- ✓ Designing handy, economic and compact hydrogen production system to overcome the storage problem.

Other devices

Organic Light Emitting Diodes (OLEDs)

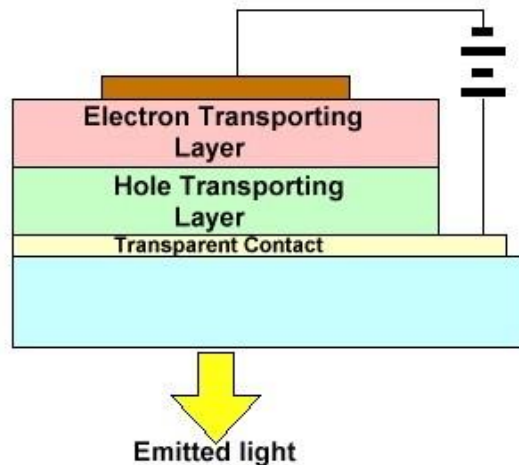
Organic light emitting diodes (devices) or OLEDs are monolithic, solid-state devices that typically consist of a series of organic thin films sandwiched between two thin-film conductive electrodes.



WHAT IS AN OLED?

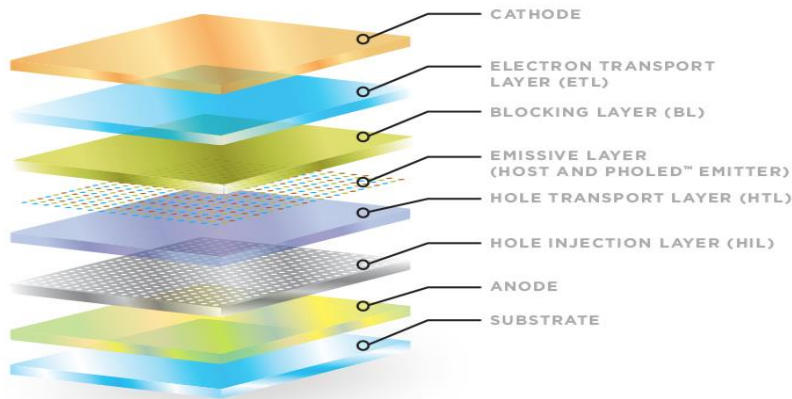
- OLEDs are solid state devices composed of thin films of organic molecules that is 100 to 500 nanometres thick.
- They emit light with the application of electricity.
- They don't require any backlight. i.e., they are self emitting.
- They are made from carbon and hydrogen.

OLED Structure



HISTORY

- The first OLED device was developed by Eastman Kodak in 1987.
- In 1996, pioneer produces the world's first commercial PMOLED.
- In 2000, many companies like Motorola, LG etc developed various displays.
- In 2001, Sony developed world's largest fullcolor OLED.



The basic OLED cell structure consists of a stack of thin organic layers sandwiched between a conducting anode and a conducting cathode.

- Substrate (can be plastic, glass, or metal foil) – Foundation of the OLED
- Anode (may or may not be transparent depending on the type of OLED) – Positively charged to inject holes (absence of electrons) into the organic layers that make up the OLED device
- Hole Injection Layer (HIL) – Deposited on top of the anode this layer receives holes from the anode and injects them deeper into the device
- Hole Transport Layer (HTL) – This layer supports the transport of holes across it so they can reach the emissive layer
- Emissive Layer – The heart of the device and where light is made, the emissive layer consists of a color defining emitter doped into a host. This is the layer where the electrical energy is directly converted into light.
- Blocking layer (BL) – Commonly used to improve OLED technology by confining electrons (charge carriers) to the emissive layer
- Electron Transport Layer (ETL) – Supports the transport of electrons across it so they can reach the emissive layer.
- Cathode (may or may not be transparent depending on the type of OLED) – Negatively charged to inject electrons into the organic layers that make up the OLED device.

WORKING PRINCIPLE

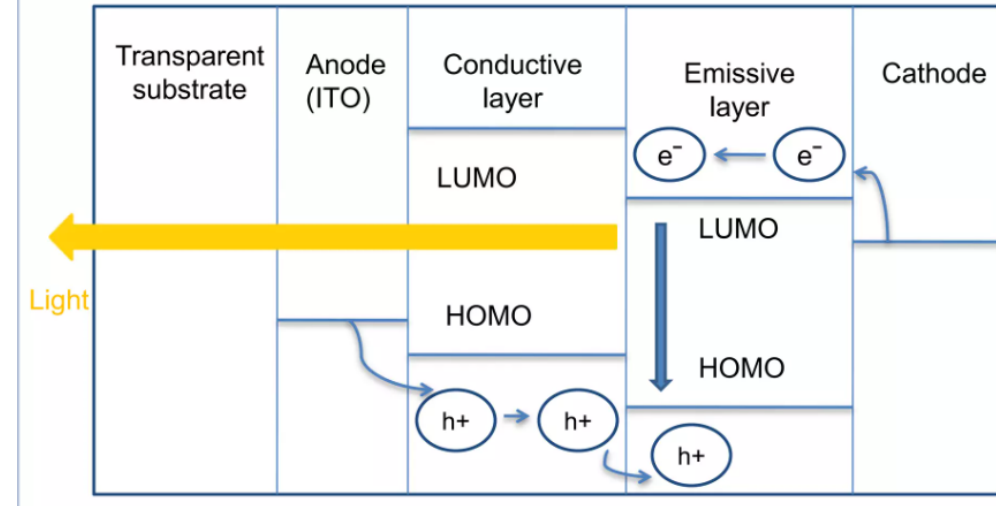
- A voltage is applied across the anode and cathode.
- Current flows from Anode to cathode through the organic layers.
- Electrons flow to emissive layer from the cathode.
- Electrons are removed from conductive layer leaving holes.
- Holes jump into emissive layer.
- Electron and hole combine and light emitted.

TYPES OF OLED

Six types of OLEDs

- Passive matrix OLED(PMOLED).
- Active matrix OLED(AMOLED).
- Transparent OLED(TOLED).
- Top emitting OLED.
- Flexible OLED(FOLED).
- White OLED(WOLED).

OLED DEVICE OPERATION



OLED vs. LCD

OLED	LCD
○ Greater view angle. ○ High contrast. ○ Faster response time. ○ Do not require backlighting. ○ Temperature($\sim 50^{\circ}\text{C}$ – 80°C).	○ Limited view angle. ○ Low contrast. ○ Slow response time. ○ Require backlighting. ○ Temperature($\sim 0^{\circ}\text{C}$- 100°C).

Thank you